Progenitors

The DICTO Hecatodihedron and the DICTO Ogdoecontahedron as a twin pair: on their own they never close golden; set into the Spheres of Dharma in place of the dodecahedron they close it again on φ^9, and the cycle then holds the core and hull of two dual compounds



DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
Version 1 · 16 October 2026 · DICTO DOI: 10.5281/zenodo.23413184 (all versions) · Licence CC BY 4.0 · Data, scripts, exact checks and renders: https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-progenitors)
The idea of the two solids as the progenitors of a new cycle, the truncated icosahedron's place and all names are DICTO's; the exact computations, searches and text were done with an AI assistant (Claude, Anthropic); a second model (Fable) advised on scope and publishing order.

Abstract. The DICTO Hecatodihedron H (102 faces) and its polar, the DICTO Ogdoecontahedron O (80 faces), came out of the Spheres of Dharma, an exact cycle of icosahedral shells that repeats every φ^3. DICTO asked whether the twin pair could seed a cycle of its own. Every solid here is exact in $\mathbb{Q}(\sqrt{5})$, and every nesting and volume is checked with remainder 0. On their own the twins do not close golden. H, O and their classical relatives, nested tightly in every order (8,742 cycles), never return on a power of φ, and the tightest 3-cycle gains exactly $9/\varphi^4 = (3/\varphi^2)^2$ per round: a cycle governed by 3, as in *Double Vision*. Set into the Spheres of Dharma with O in the dodecahedron's place, they close it again on φ^9 in volume: I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ^3. The cycle then holds the core and hull of two dual compounds, the icosahedron–dodecahedron compound (ID, RT) and the twins' own compound, the DICTO Hecatogdoecontahedron (rTI, jtI). Prior-art status is stated claim by claim in §4.

1. The question

The Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833) nest six icosahedral solids, icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ^3 larger, each layer filled exactly by one kind of piece. Its in-between stage between the truncated icosahedron and the triacontahedron is the DICTO Truncated-Icosahedral Hecatodihedron H (DICTO DOI: 10.5281/zenodo.23360857): 12 pentagons, 30 rhombi and 60 triangles, the joined truncated icosahedron jtI with its 12 five-fold corners cut flat (Version 2 of that paper). Its polar is the DICTO Ogdoecontahedron O (DICTO DOI: 10.5281/zenodo.23404688): 20 regular hexagons and 60 kites, the canonical pentakis dodecahedron with its 20 three-fold corners cut flat. Both are derived from classical solids; neither is in the classical catalogues.

DICTO's question: let H and its dual be the *progenitors* of a new cycle. Can they nest into an exact, endlessly repeating cycle of their own, and does it close on a power of φ, as the Spheres of Dharma do? We write I, ID, D, TI, PD, RT for the icosahedron, icosidodecahedron, dodecahedron, truncated icosahedron, pentakis dodecahedron and rhombic triacontahedron, all in the Dharma's sizes (icosahedron edge $1/\varphi$). Figure 1 shows the answer found in §3.

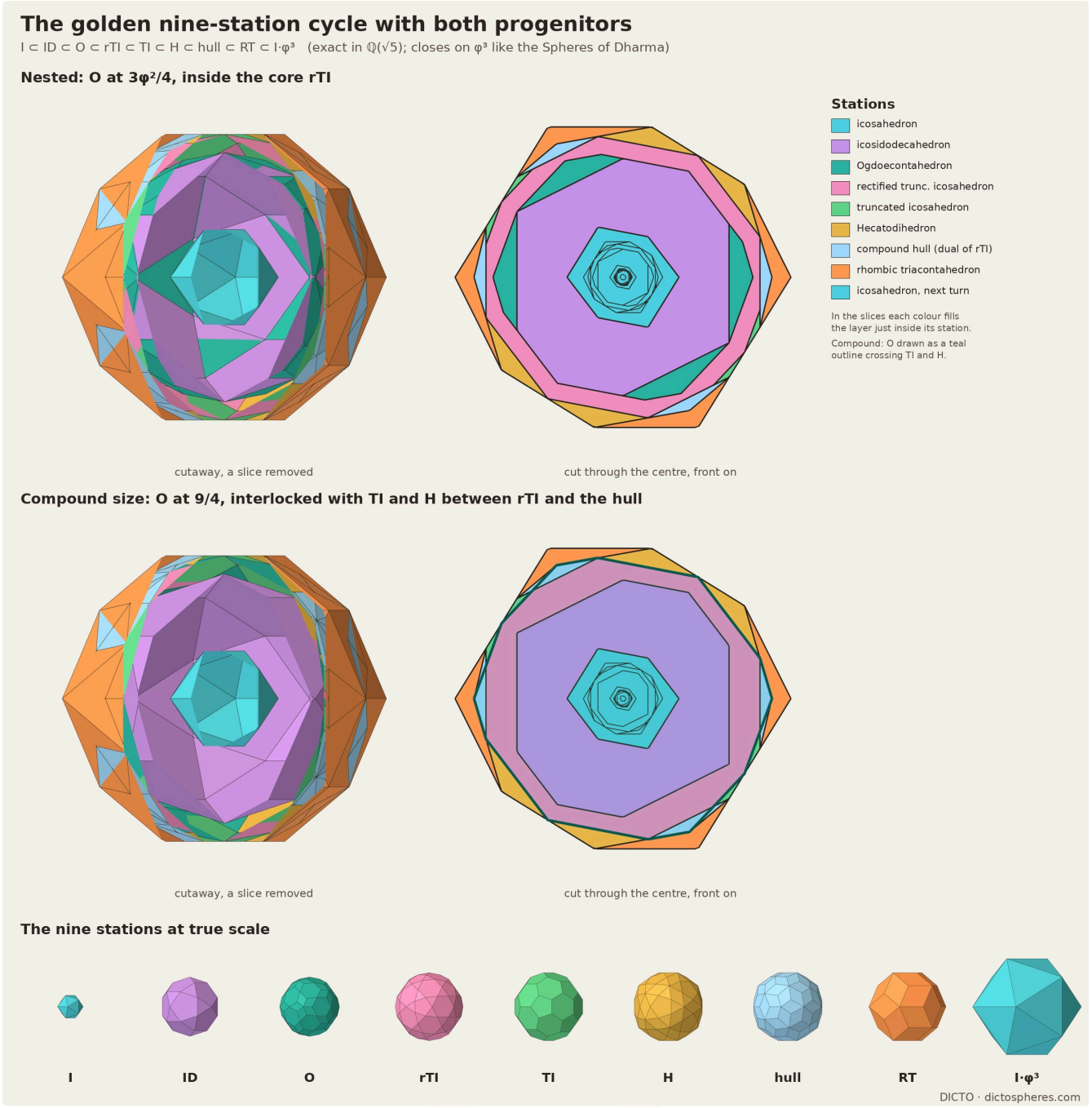


Figure 1. The golden cycle with both progenitors: I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ hull (jtI) ⊂ RT ⊂ I·φ³. Top: O at 3φ²/4, nested inside the core rTI, cutaway and a front-on cut through the centre. Middle: O at its compound size 9/4, interlocked with TI and H. Bottom: the nine stations at true scale, and the H–O compound with its common core and hull. Drawn from exact stations.

2. On their own: no golden closure

The pair. H and O nest endlessly in each other, but the round trip does not close golden. H ⊂ s·O with s = 81/8 − 27√5/8 (20 corners touch), and O ⊂ t·H with t = (170 + 2√5)/361 (12 touch). The round trip is s·t = (3375 − 1107√5)/722 ≈ 1.246, which is not a power of φ.

The quartet. Adding the truncated icosahedron (DICTO: "maybe the truncated icosahedron has a place") and its dual PD gives four solids: TI ⊂ H and O ⊂ PD touch fully, and TI ↔ PD, H ↔ O are dual pairs. Taken in every cyclic order at the least size that holds the previous solid, none of the six orders closes on a power of φ. The tightest, TI ⊂ H ⊂ O ⊂ PD ⊂ TI, gains exactly (63 − 27√5)/2 = 9/φ⁴ = (3/φ²)² per round (exact). That is an endless cycle governed by

3, not by φ. It is the same factor 3 that keeps the two toggle settings of *Double Vision* apart (DICTO DOI: 10.5281/zenodo.23407106).

A wider search. All 8,742 cycles through TI, H, O and PD, together with any of I, ID, D and RT (each at most once, in any order), were tried. Each solid was set at the least size that holds the previous one, and none closes on a power of φ (exact). The criterion is strict: every step is tight. The Spheres of Dharma themselves fail it, since two of their steps, I → ID and RT → I·φ³, are loose. A golden cycle through the progenitors must therefore allow loose steps.

3. A golden cycle with both progenitors (exact)

The icosidodecahedron is a common core of both progenitors. O wraps it, with its 30 corners on O's faces and each triangle of ID inside a hexagon of O. Theaetetus grows it through Cleo-Icos to H (DICTO DOI: 10.5281/zenodo.23378070). Placing O on the Dharma's own ID, at scale 3φ²/4 (the middle setting of *Double Vision*), gives a cycle in which every step nests exactly:

$$\mathbf{I \subset ID \subset O \subset rTI \subset TI \subset H \subset jtI \subset RT \subset I \cdot \varphi^3}$$

The closure is I·φ³ / I = 38 + 17√5 = φ⁹ in volume, golden as in the Spheres of Dharma. Here rTI is the rectified truncated icosahedron and jtI its dual, the joined truncated icosahedron. Table 1 lists the contacts. O and D cross, so O takes D's place; they are alternatives, as H and 3·I are in the Dharma.

Step	Contact (exact)
I ⊂ ID	loose (0 of 12 corners on ID's faces)
ID ⊂ O	tight: all 30 corners of ID on faces of O
O ⊂ rTI	loose
rTI ⊂ TI	tight: all 90 corners of rTI on TI's surface (its edges)
TI ⊂ H	tight: all 60 corners of TI on faces of H
H ⊂ jtI	tight: all 80 corners of H on faces of jtI
jtI ⊂ RT	tight: 80 corners of jtI on faces of RT
RT ⊂ I·φ³	loose (0 of 32 corners on faces of I·φ³)

Table 1. The nine-station cycle (exact). O at 3φ²/4; rTI, the core of the H–O compound, at the size it has in the truncated icosahedron's midsphere.

Both compounds in one cycle. The cycle now holds the core and hull of two dual compounds. ID and RT are the core and hull of the icosahedron–dodecahedron compound. rTI and jtI are the core and hull of the Hecatodihedron–Ogdoecontahedron compound, which DICTO named the DICTO Hecatogdoecontahedron: H and O interlocked in the truncated icosahedron's midsphere, 182 faces and 182 corners, self-dual. Its core and hull are those of the classical truncated icosahedron–pentakis dodecahedron compound, as #28 Version 2 explains. We claim only the pairing, not the core or the hull.

4. Relation to known work, and status

- **Known, cited, not claimed.** Nested Platonic solids go back to Kepler (1596), and nested icosahedral shells appear in virology and fullerene work (Dechant et al. 2014). The truncated icosahedron–pentakis dodecahedron compound, the rectified truncated

icosahedron and its dual jtI are classical (MathWorld; Wikipedia; McCooey; Hart). H is the jtI with its five-fold corners cut flat and O the canonical pentakis dodecahedron with its three-fold corners cut flat (the Version 2 papers). Polar reciprocation is classical.

- **Not found, in the sources checked:** a golden cycle containing H and O, or one holding the core and hull of both compounds; the tight-nesting results of §2. The name "Hecatogdoecontahedron" returns no results.
- **Searched (16 October 2026).** Wikipedia, MathWorld, the Polytope Wiki, Klitzing, OEIS, McCooey's Visual Polyhedra, G. Hart's *Virtual Polyhedra* (index, compound list and the full alphabetic model list: compounds of the uniform solids with their duals, the rhombic enneacontahedron, the pentakis dodecahedron; none of the solids or cycles here), Wenninger's model lists, arXiv (nested golden shells, polar sequences), and the DICTO reference library.
- **Antiprism** (A. Rossiter, version 0.29), run 16 October 2026: it recomputes the solids independently, and a sweep of every Conway form with up to two operators on I and D (840 forms) matches neither H nor O, while the H–O core and hull have the structures of atI and jtI.
- **Not read.** Wenninger's *Polyhedron Models* and *Dual Models* in print; Coxeter, Cromwell and Holden in print. The Stella catalogue was not run. Every "not found" above is limited to the sources listed; the full log is in the study (SEARCH-LOG.md).

5. Conclusion

The twins do not make a golden cycle on their own. Tight nestings of H, O and their classical relatives are governed by the factor 3, as in *Double Vision*, and never close on a power of φ. Set into the Spheres of Dharma, with O in the dodecahedron's place, they close golden at once, on φ^9 as the Dharma itself does. The cycle then holds the core and hull of both dual compounds, the classical icosahedron–dodecahedron one and DICTO's own. So the progenitors do not start a new cycle; they give the old one a second reading in which the dodecahedron's place is taken by its non-classical cousin. A companion paper takes up the question this leaves: where the dodecahedron goes, and what fills the cycle's two loose steps. In brief, three more stations close them, and the cycle grows to twelve: I ⊂ Cleo star ⊂ D′ ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ M ⊂ I·φ^3. Here D′ is the dodecahedron at two thirds, the hull of DICTO's Cleo star, and M is the DICTO Metaxyhexecontahedron. **Open:** a golden cycle through the twins with loose steps of a different kind, and a physical build of the nine stations.

6. Reproducibility

In studies/2026-10-16-progenitors: `pair_chain.py`, `quartet.py`, `golden_search.py` (§2); `o_dharma.py`, `oh_compound.py`, `core_hull_in_cycle.py` (§3); the Antiprism files in `antiprism/`. Exact arithmetic in $\mathbb{Q}(\sqrt{5})$ (numbers a + b√5, a and b rational), with exact sign tests for every inclusion. Figure 1 from `render_nine.html`. The full record, with every step and its corrections, is in the study README and SEARCH-LOG.md.

References

1. DICTO, Spheres of Dharma: a second endless golden cycle of the icosahedral solids, through the icosidodecahedron, the fullerene cage and the rhombic triacontahedron, DICTO DOI: 10.5281/zenodo.23310833.

2. DICTO, The DICTO Truncated-Icosahedral Hecatodihedron (Version 2), DICTO DOI: 10.5281/zenodo.23360857.
3. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
4. DICTO, Inverted Shells: The Spheres of Dharma Turned Inside Out, DICTO DOI: 10.5281/zenodo.23405029.
5. DICTO, Double Vision – Both Cycles in One Space, DICTO DOI: 10.5281/zenodo.23407106.
6. DICTO, Theaetetus: an icosahedron held exactly by its Cleo-Icos, grown to the DICTO Hecatodihedron in two exact stories, DICTO DOI: 10.5281/zenodo.23378070.
7. J. Kepler, *Mysterium cosmographicum*, Tübingen, 1596.
8. J. H. Conway, H. Burgiel, C. Goodman-Strauss, *The Symmetries of Things*, A K Peters (2008) (Conway polyhedron notation).
9. E. W. Weisstein, Pentakis Dodecahedron; Truncated Icosahedron–Pentakis Dodecahedron Compound, MathWorld, https://mathworld.wolfram.com.
10. D. McCooey, Visual Polyhedra: Archimedean–Catalan Hulls, Rectified Archimedean Solids, Biscribed solids, https://dmccooey.com/polyhedra (accessed 16 October 2026).
11. Rhombic enneacontahedron (Conway jtI, dual of the rectified truncated icosahedron), Wikipedia, https://en.wikipedia.org/wiki/Rhombic_enneacontahedron; Rectified truncated icosahedron, https://en.wikipedia.org/wiki/Rectified_truncated_icosahedron.
12. G. W. Hart, *Virtual Polyhedra*, https://www.georgehart.com/virtual-polyhedra/vp.html (alphabetic list read 16 October 2026).
13. A. Rossiter, Antiprism, polyhedron modelling software, version 0.29, https://www.antiprism.com.
14. P.-P. Dechant, J. Wardman, T. Keef, R. Twarock, Viruses and fullerenes: symmetry as a common thread?, *Acta Cryst. A* 70 (2014) 162–167, doi:10.1107/S2053273313034220.
15. M. J. Wenninger, *Polyhedron Models*, Cambridge University Press (1971); *Dual Models*, Cambridge University Press (1983) (not read; see §4).